

METEOR: From Raw Surround-View Recordings to a Seven-Task Driving Network without Human Labels or Human-Written Code

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<https://github.com/tier4/METEOR>

Abstract

We present METEOR (Multi-task Estimation of Traffic Elements, Objects and Roads), a surround-view, camera-only network that predicts seven driving tasks in a single forward pass—BEV lane segmentation, metric depth, oriented 3D boxes, 21-class 2D segmentation, 10-class 2D detection, 3D semantic occupancy, and an end-to-end driving output (trajectory, steering, acceleration, braking)—and, equally importantly, the fully automated pipeline that builds it. Every supervision signal is distilled offline from raw eight-camera recordings of the Co-MLOps Data Recording System by an autolabel factory built on the CoMET autolabeling foundation, which combines LiDAR, ego-motion and 2D panoptic pseudo-labels; no human annotated a single box, mask, or trajectory. The network couples a shared image backbone with a *depth-gated inverse perspective mapping* that uses predicted depth as a visibility gate for geometric BEV projection, and restricts itself to TensorRT-exportable operators (38.3 M parameters, 3.1 TFLOPs for eight cameras). We describe a rolling-round training methodology in which data conversion, ground-truth refinement and multi-task training run concurrently, and report a series of measurement-driven loss corrections—curvature-weighted imitation, coverage-preserving thin-class rasterisation, and small-object retention—each of which repaired a quantified failure mode. On a held-out recording, METEOR reaches 0.292 BEV lane mIoU, 1.60 m depth MAE, and a 3 s trajectory ADE of 0.76 m (0.72 m on curves, brought down from 2.31 m by curvature weighting). Finally, the entire system—models, pipeline, training and analysis code, and this manuscript—was authored by a large language model agent (Claude Fable 5) operating autonomously under human direction, which we discuss as a development methodology.

1 Introduction

Perception stacks for automated driving are conventionally built one task at a time, each with its own annotation campaign. Labeling is the dominant cost: boxes, masks, lane geometry, and occupancy grids each require dedicated tooling and thousands of hours of human effort, and every new task restarts the cycle. At the same time, modern data-collection platforms record far more information than any annotation budget can absorb—synchronized surround cameras, LiDAR, and centimeter-level ego-motion—which in principle contain

almost everything the labels would say.

This paper describes an attempt to close that loop completely. Starting from raw recordings of the Co-MLOps Data Recording System (DRS)¹—eight synchronized cameras, a concatenated LiDAR, ego-poses, and machine-generated 2D panoptic labels—we build *seven* supervision signals and train a single multi-task network, with two hard constraints that we believe make the result practically interesting:

- **Zero human labels.** All ground truth is produced by an autolabel factory built on CoMET, the autolabeling foundation of the Co-MLOps project. Humans never annotate; they only record.
- **Zero human-written code.** The complete system—ground-truth extractors, model, trainer, demo tooling, documentation, and this paper—was written by an LLM agent (Claude Fable 5) working autonomously under human direction (Sec. 7).

Our contributions are:

1. **An autolabel factory for seven tasks** (Sec. 3): a ten-stage, per-scene, resumable pipeline that turns raw DRS recordings into BEV lane maps, dense metric depth, camera-confirmed oriented 3D boxes, 21-class 2D segmentation, 10-class 2D boxes, end-to-end driving targets derived purely from ego-motion, and 3D semantic occupancy with ray-carved free space—plus the quality gates that make such labels trainable.
2. **METEOR** (Sec. 4): a 38.3 M-parameter, 3.1 TFLOP surround-view network whose depth-gated IPM converts image features to BEV using only TensorRT-friendly operators, and whose head design follows an explicit capacity rule—*parameters at low resolution, compute at high resolution*—that adds tasks at < 5% marginal cost.
3. **A rolling-round training methodology** (Sec. 5): data conversion, GT re-extraction and training run concurrently; each ~24 h round folds in newly converted scenes and measurement-driven loss fixes. We document three such fixes (curvature weighting for imitation, coverage-based thin-class rasterisation, small-object retention) with the quantified failures that motivated them.
4. **An agent-authored system** (Sec. 7): to our knowledge the first published driving-perception system in which

¹<https://co-mlops.tier4.jp/>

not only the labels but all engineering artefacts were machine-generated, with humans reduced to specification and review.

2 Related Work

Backbones and dense prediction. Our encoder follows the standard lineage of convolutional backbones [1–4] with batch normalisation [9] and multi-scale fusion by FPN [8]; transformer encoders [5, 6] and high-resolution networks [7] are attractive but conflict with our strict deployment constraint. Dense prediction heads descend from FCN [10] and its successors [11–14], with efficiency-oriented designs [15, 16] closest to our encoder–decoder segmentation head. Panoptic labels [17, 18] in COCO RLE format [19] are the raw material of our factory, and the class systems echo urban-scene benchmarks [20, 21].

2D object detection. Two-stage [22, 23] and one-stage [24–29] detectors are both well explored; keypoint-based formulations [30, 31] and set prediction [32, 33] removed anchors and NMS respectively. Our 2D head is a three-scale CenterNet [31]; YOLO-style multi-scale assignment [26] with heatmap decoding that reduces to a max-pool, which survives TensorRT export.

Lane and map perception. Image-space lane detectors evolved from spatial message passing [34] through row-anchor and line-proposal methods [35–38] and polynomial regression [39]; 3D lane benchmarks [40–42] moved the task to metric space. Online HD-map construction rasterises or vectorises BEV lanes [43–48]; our factory produces exactly such vectorised lane GT—but offline, from LiDAR accumulation rather than human annotation, in the spirit of classic skeleton-and-simplify cartography [143, 144].

Camera-to-BEV lifting and 3D detection. Splatting a predicted depth distribution [49] underlies a large family of camera-BEV detectors [50–53]; query- and attention-based alternatives include DETR3D [54], PETR [55, 56], BEVFormer [57], sparse recurrent variants [58, 59], and cross-view transformers [60]; simpler bilinear-sampling baselines are surprisingly strong [61–63]. Monocular baselines [64, 65] established camera-only 3D detection. Our depth-gated IPM is a pull-based hybrid: like classic IPM [66, 67] it samples at geometric ground intersections (including fisheye models [68]), but the sample is gated by the predicted depth probability at the ray range—LSS run backwards, with `gather` instead of `scatter`. LiDAR detectors [69–74] and fusion methods [75, 76] inform our CenterPoint-style BEV head, which we supervise with camera-confirmed LiDAR annotations only.

Occupancy. Semantic scene completion began indoors [77] and on LiDAR [78]; camera-based 3D occupancy is now a benchmark task [79–86]. Our contribution is on the label side: multi-sweep labeled-LiDAR voxelisation with dynamic-object de-smearing and dense ray-carved free space, produced without any human input.

Depth estimation. Supervised monocular depth spans direct regression [87], ordinal and bin formulations [88–90], self-supervised photometric training [91], and large-scale generalist models [92, 93]. Our depth GT densification is

closer to classical completion [94, 95], but segment-aware: interpolation runs inside merged panoptic road segments.

End-to-end driving. From ALVINN [96] and PilotNet [97] through conditional imitation [98, 99], driving-by-planning hybrids [100, 101], simulator-centric agents [102–107], and BEV-planning stacks [108–110], imitation from ego-motion is a long-standing recipe. Our E2E head is deliberately minimal (speed-conditioned regression on the shared BEV feature); its interest lies in the free supervision and in the documented imbalance corrections, evaluated with curve-conditioned metrics on real recordings [112, 113].

Datasets and auto-labeling. Public driving corpora [111, 112, 114–117] standardised the schema our recordings follow. Offboard auto-labeling [118, 119] distills strong detectors into labels; the broader semi-/self-supervised toolbox [120–126] motivates training on machine-generated targets. Our factory is detector-free for five of seven tasks and covers modalities (lanes, depth, occupancy, driving) that offboard detection pipelines do not.

Multi-task learning. Sharing one trunk across tasks is classical [127–129]; loss balancing has been addressed by uncertainty weighting [130], gradient normalisation [131], and gradient surgery [132]. We use fixed hand-tuned weights corrected by measurement (Sec. 5.2); with seven tasks and rolling data we found interpretable static weights easier to operate than adaptive schemes.

LLM agents for engineering. Code generation has progressed from benchmark-level synthesis [145, 146] to tool-using agents [147] evaluated on real repository tasks [148] with rapidly improving frontier models [149]. This project is, to our knowledge, an early data point on a full ML *system* (not a patch or a script) built end-to-end by such an agent, including its own measurement loop (Sec. 7).

3 The Autolabel Factory

The factory consumes t4dataset-format recordings (a nuScenes-like schema [112]: keyframed samples, calibrated sensors, ego-poses, LiDAR sweeps, and 2D panoptic annotations stored as COCO run-length masks). A single driver executes ten stages per scene; every stage skips existing outputs, so the factory is resumable and runs scene-parallel alongside training. We summarise each supervision signal; exact recipes are in the repository documentation.

3.1 BEV lane ground truth

LiDAR points from every keyframe are projected into all cameras (pinhole and equidistant fisheye [68]) and take road-surface classes from the panoptic masks; thin classes (lane line, stop line, road edge, marking) are only accepted within 15 m, where the pixel-to-ground error is sub-cell. Ego-poses accumulate the labeled points into a global 0.1 m grid over the whole scene, which is rasterised with hit-count priorities and a 20 m observation-distance mask. The raster is then *vectorised*—skeletonisation [144], spur pruning, Douglas-Peucker simplification [143], and dash-gap linking—and per-frame GT crops ($\pm 80 \times \pm 50$ m at 0.2 m) are re-rendered hybrid-style: area classes from the raster, line classes re-

drawn from the vectors at fixed width. Two guards proved essential: enclosed road holes (≤ 400 cells, LiDAR shadows) are filled, and opposing carriageways are removed by an ego-connected-component filter with a 30% revert guard that protects wide intersections from being wiped.

3.2 Dense metric depth

Per camera, LiDAR points splat a sparse min-depth image at stride 4. Depth is densified—in the spirit of classical completion [94, 95]—by linear interpolation *over the union of road and road-paint panoptic segments*—interpolating paint separately leaves holes around every lane line—followed by a geometric ground-plane fill restricted to rays within 50° of downward (a guard against fisheye-calibration tails), a same-segment nearest fill, and per-segment medians. Sky and ego pixels receive fixed codes. The result is a 64-bin (1.25 m) target at 108×192 for all eight cameras.

3.3 Camera-confirmed 3D boxes

The recordings carry LiDAR-frame 3D annotations whose instance IDs are disjoint from the 2D panoptic instances, so association must be geometric: a 3D box is kept only if its eight projected corners overlap a same-class 2D annotation in some camera (intersection over minimum area > 0.3). This removes fully occluded objects that a camera-only network cannot see, which otherwise poison the BEV detection loss.

3.4 2D segmentation and 2D boxes

Both taxonomies are CSV-driven so that class definitions are data, not code: 21 semantic classes and 10 instance classes with Cityscapes-like colours. Two findings matter. (i) *Thin classes do not survive naive downsampling*: at stride-4 GT resolution a lane line is ~ 0.3 px wide and nearest-neighbour resizing deletes it; rasterising a cell as the thin class when $> 12\%$ of its footprint is covered recovered $+49\%$ lane pixels and turned broken dashes into connected supervision. (ii) *Silent truncation deletes exactly the rare classes*: with a per-camera box budget of 32, area-sorted retention dropped 29% of annotations in crowded scenes—all of them small (cones, traffic lights, distant unknowns). Raising the budget to 96 and adding per-class positive weights fixed the small-object classes.

3.5 End-to-end driving targets from ego-motion

No CAN bus data is available, so all driving targets derive from ego-poses: future positions at $+0.5 \dots + 3.0$ s transformed into the current ego frame (six waypoints); speed and acceleration by smoothed finite differences; steering via a bicycle model $\delta = \arctan(L\dot{\psi}/v)$ with wheelbase $L=2.8$ m, masked below 0.5 m/s; braking as acceleration below -0.5 m/s². Instantaneous signals are written for *every* frame—an early version left the speed field zero when the 3 s future was missing at scene tails, and the deployed model, conditioned on that bogus zero, predicted “hold” at 80 km/h. Trajectory targets carry a validity mask instead.

3.6 3D semantic occupancy

Ego-frame voxels ($16 \times 200 \times 200$ at 0.4 m; ± 40 m, $z \in [-1, 5.4)$ m) with ten classes plus *unknown*. Points from ± 8 strided frames are labeled by sampling the cached 21-class segmentation GT (a factor-15 speedup over re-decoding panoptic masks) and accumulated via ego-poses; *dynamic* classes (vehicles, two-wheelers, pedestrians) are taken from ± 1 frames only, since full accumulation smears moving objects into lane-long streaks. Free space is carved by ray-stepping the current sweep at 0.4 m intervals; coarser stepping (an early 1.7 m version) left most above-road voxels unknown, and the trained model hallucinated confident structure in exactly those unsupervised regions.

3.7 Quality gates

Three train-time filters removed systematic label noise: scene-end trimming (the last 10 frames lack forward accumulation), a per-frame GT-coverage filter (labeled fraction $\geq 3\%$ in the ± 30 m band and $\geq 0.5\%$ in the 30–80 m band; long-stationary frames fail this), and indoor/underground scene rejection via panoptic sky statistics, because NDT [142] pose drift makes kinematic filters unreliable there.

4 METEOR Architecture

Fig. 1 shows the network. Eight 768×432 images pass through a shared ResNet-34 [2] with an FPN [8] that fuses all levels at stride 4 into a 160-channel feature f .

4.1 Depth-gated IPM

A fixed BEV grid (800×500 at 0.2 m; $\pm 80 \times \pm 50$ m, $z=0$) is projected into every camera by the calibrated K, T . At each projection we `grid_sample` a 96-channel context feature c and the predicted depth distribution $p(d)$, then `gather` the probability at the point’s geometric range r :

$$w_{i,n} = p_n(d=r_{i,n}) + \epsilon, \quad b_i = \frac{\sum_n w_{i,n} c_n(u_{i,n})}{\sum_n w_{i,n}}, \quad (1)$$

where n indexes cameras, i BEV cells, and $\epsilon=0.05$ keeps the gate leaky. Depth thus acts as a *visibility valve*: features flow into a BEV cell only from cameras whose predicted depth agrees with that cell’s range, suppressing the occlusion bleed-through that plagues plain IPM. The block has no parameters and uses only `grid_sample/gather`, in contrast to scatter-based splatting [49] or attention [57].

4.2 Heads and capacity allocation

Table 1 lists all stages. Two rules organise the design:

Parameters at low resolution, compute at high resolution. Early heads ran 3×3 convolutions at stride 4—the worst corner of the FLOPs/params trade-off. The redesigned 2D heads push channels into stride-8/16 towers (encoder-decoder for segmentation; a three-scale CenterNet pyramid for detection, with GT assigned by box size) and touch stride

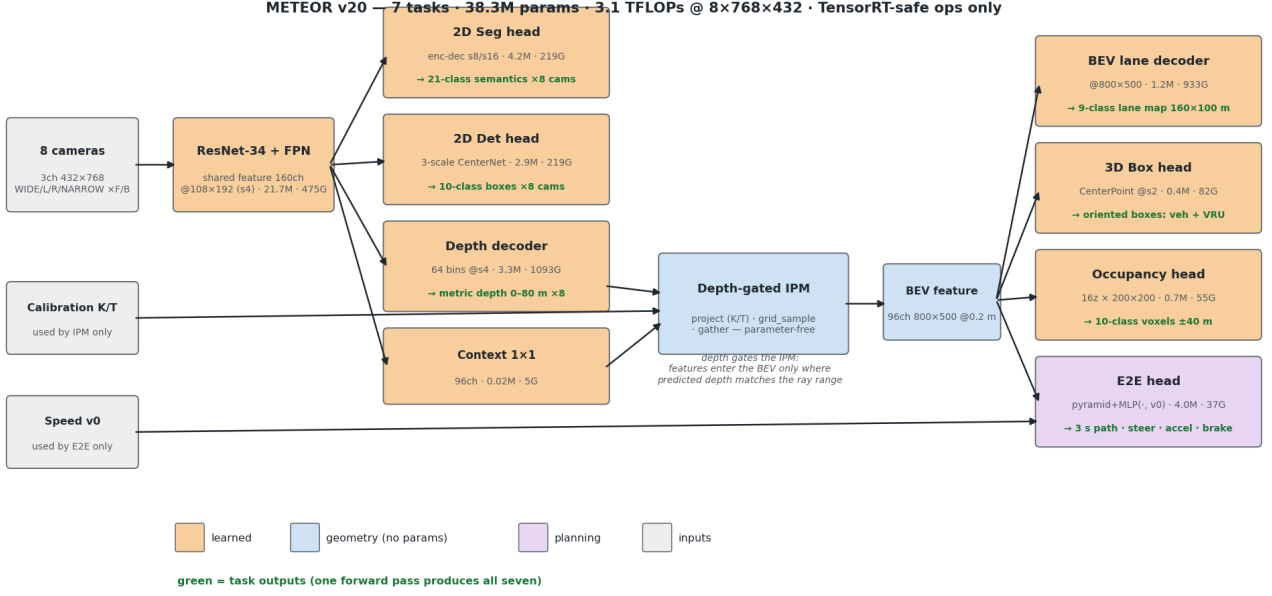


Figure 1: METEOR architecture. A shared ResNet-34+FPN yields a stride-4 feature per camera, consumed by four image-space heads. The parameter-free depth-gated IPM projects BEV grid points into the cameras and gathers context features weighted by the predicted depth probability at each point’s range; four BEV-space heads (lanes, 3D boxes, occupancy, E2E) share the resulting 96-channel BEV feature. Green lines denote task outputs; all operators are TensorRT-exportable.

Table 1: Measured cost per stage (eight cameras, 768×432).

Stage	Params	GFLOPs	Share
Backbone + FPN	21.66 M	475	15%
Depth decoder	3.29 M	1093	35%
BEV lane decoder	1.16 M	933	30%
2D segmentation head	4.18 M	219	7%
2D detection head	2.85 M	219	7%
3D box head	0.41 M	82	3%
Occupancy head	0.70 M	55	2%
E2E head	4.04 M	37	1%
Context + IPM	0.02 M	6	<1%
Total	38.31 M	~3118	

4 only through 1×1 laterals: an 18× parameter increase for ~2× FLOPs, worth +0.076 2D-seg mIoU (Sec. 6.4).

Tasks are cheap on a shared BEV. The occupancy head (a stride-2 stem on a ±40 m crop predicting 10×16 channels) and the E2E head (a strided pyramid pooled to 256-d, concatenated with speed v_0 , and decoded by an MLP) together cost 3% of total compute.

5 Training

5.1 Multi-task loss

The total loss is a weighted sum over tasks: BEV lanes (class-weighted cross-entropy with trained background, Lovász-softmax [138], boundary re-weighting, Tversky [139] on line classes, Dice [140] on crosswalks, and far-row weighting), depth (label-smoothed [136] bin cross-entropy plus an L1 penalty on expected metric depth), 3D boxes (Gaussian

focal [27, 31] + centre L1; IoU-based box losses [141] were unnecessary at our grid resolution), 2D segmentation and detection (class- and camera-weighted variants), occupancy (class-weighted cross-entropy ignoring unknown voxels), and E2E (below). Aux losses return a graph-preserving zero on all-ignore batches—during rolling GT re-extraction whole batches can be unlabeled, and an unguarded mean cross-entropy returns NaN and destroys training within 50 steps.

5.2 Measurement-driven loss corrections

Three corrections were made *after* quantifying a failure, and we report them as results in Sec. 6: (i) **curvature weighting**: 83% of frames are near-straight and longitudinal waypoint magnitudes exceed lateral ones 18×, so plain L1 learns a go-straight prior; we weight lateral errors 4× and scale each sample by $1 + \min(|y_{3s}|, 6)/1.5$. (ii) **coverage-based thin-class rasterisation** and (iii) **small-object retention** as described in Sec. 3.4, paired with per-class positive weights in both detection heads and a near-range (<20 m) positive boost for vulnerable road users in the 3D head.

5.3 Rolling rounds

Training proceeds in ~24 h rounds on seven GPUs. Each round refreshes the scene list with newly converted recordings, warm-starts from the previous best checkpoint (shape-mismatched heads are dropped and re-initialised, so class-count changes retrain only the affected head), and applies whatever GT or loss fixes the previous round’s metrics motivated. Nine rounds and four new task heads shipped in the project’s first three days; the same loop is designed to run for months as the factory converts the full corpus.

Table 2: Validation results on a held-out recording.

Metric	Value
BEV lane mIoU (8 classes + bg)	0.292
2D segmentation mIoU (21 classes)	0.451
lane line / pole IoU	0.483 / 0.449
Depth MAE (0–80 m, 8 cams)	1.60 m
E2E trajectory ADE / FDE (3 s)	0.76 m / 1.65 m
E2E curve ADE ($ y_{3s} > 2$ m)	0.72 m
Steering MAE / brake accuracy	0.027 rad / 0.81
Occupancy road IoU (0–10 / 10–20 / 20–40 m)	0.63 / 0.64 / 0.64

6 Experiments

6.1 Setup

Data: $\sim 2,000$ scenes from two vehicle platforms (1,120 from an initial corpus; a second 5,282-scene corpus is converted progressively and folded in round by round). Validation is one held-out recording session (273 scenes) never used for training. Training: $7\times$ data-parallel, batch 2 per GPU, AdamW [133, 134], cosine schedule [135], 10^{-4} peak LR, 46k samples per epoch. All numbers below are single-model, single-pass; no test-time augmentation.

6.2 Main results

Table 2 summarises the current operating point. We stress the context: every number is achieved *without a single human label*, and the validation recording is from a different day and route than all training data. Fig. 2 shows qualitative output.

6.3 Curvature weighting for imitation

Binning validation frames by ground-truth lateral displacement at 3 s exposed the go-straight prior of the unweighted model: ADE was 0.62 m on straight frames but 1.30 m at 2–4 m lateral and **2.29 m** beyond 4 m— $3.7\times$ worse than straight, hidden by an aggregate ADE of 0.86 m because 76% of validation frames are straight. One round of curvature-weighted training (Sec. 5.2) brought the curve subset to **1.07 m** and a second to **0.72 m**, at no cost to straight frames (aggregate ADE improved to 0.76 m). We consider the diagnosis-metric (curve-conditioned ADE, reported every epoch) as important as the fix: without it the regression would have been invisible.

6.4 Head capacity re-balancing

Replacing the stride-4 2D heads with the low-resolution-parameter design (Sec. 4) and fixing small-object GT retention raised 2D-seg mIoU from 0.375 to **0.451** in one round (lane 0.402 $\rightarrow$ 0.483, pole 0.244 $\rightarrow$ 0.449) and enabled detection of classes that had never fired before (traffic lights, distant signs, road obstacles), at +10% total FLOPs and +37% parameters.

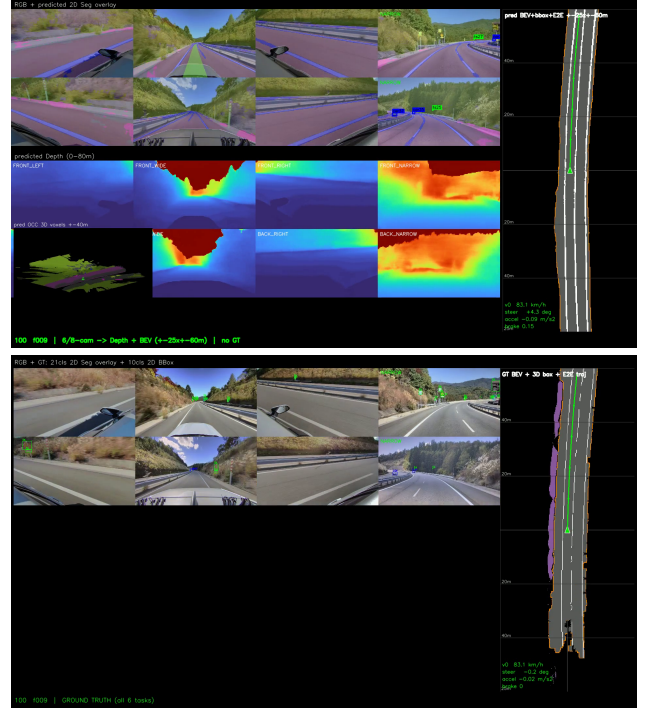


Figure 2: Qualitative results. Top: single-pass inference—eight cameras with 21-class segmentation, 2D and projected 3D boxes, and the predicted path ribbon; metric depth; predicted occupancy (isometric); BEV lanes with oriented boxes and the E2E trajectory with control gauges. Bottom: the corresponding fully automatic ground truth.

6.5 Occupancy supervision density

With 1.7 m ray steps, most above-road voxels remained *unknown* and the model hallucinated high-confidence structure there (a solid “slab” over the road in 3D renderings), while measured road IoU was unaffected—the errors live exactly where no supervision exists. Densifying carving to one sample per 0.4 m voxel multiplies free-space supervision and is expected to remove the hallucination; per-range road IoU (Table 2) already shows no near-field deficit, countering the visual impression given by top-down renderings that flatten hallucinated upper voxels onto the road.

7 Agent-Authored Development

Every artefact of this project—the ten GT extractors, twenty model variants, the trainer, the demo renderers, the documentation, and this manuscript—was written by a large language model agent (Claude Fable 5, Anthropic) operating in an autonomous loop: humans supplied goals and critiques in natural language (≈ 70 instructions over the reported period) and reviewed rendered videos and metrics; the agent designed, implemented, launched and monitored training, diagnosed failures from its own measurements, and iterated. Several of the fixes reported above originated from the agent’s own diagnostics (e.g., the KMAX truncation analysis); others from human observations that the agent then localised and repaired (e.g., the scene-tail speed bug). We do not claim the agent is autonomous in the strong sense—direction remained

human—but the division of labour (humans specify *what*, the agent implements, measures, and explains *how*) held for 100% of the code, and we found the resulting cadence (nine training rounds and four new task heads in three days) difficult to reconcile with conventional development.

8 Limitations

Thin-class BEV IoU (lane line 0.14, stop line 0.11) remains low in absolute terms; the labels themselves carry residual noise near intersections, and IoU on 1-pixel-wide classes is a harsh metric. The E2E head is pure imitation without commands or multi-modality, so intersections are resolved by data prior rather than intent. Occupancy hallucination in unknown regions is mitigated but not eliminated by denser carving; a principled treatment of camera visibility masks [85] is future work. Validation uses a single held-out recording; broader-scale evaluation across platforms awaits the full corpus conversion. Finally, the network is single-frame; temporal fusion is an obvious next step.

9 Conclusion

METEOR demonstrates that a competitive seven-task surround-view driving network can be built from raw fleet recordings with no human annotation and, in this instance, no human-written code—the autolabel factory of the Co-MLOps project (CoMET) supplies every supervision signal, and an LLM agent supplied every line of implementation. We believe the combination—data platforms that record everything, factories that label everything, agents that implement anything specified—changes the economics of adding a task from “a labeling campaign” to “a prompt and a GPU-day,” and we offer the system, its measurements, and its failure catalogue as evidence.

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